Syllabus Details

Syllabus ID:	13779
Syllabus Name:	**Game Design Fundamentals - From Concept to Creation_Thiết kế game cơ bản - Từ ý tưởng đến thực hiện**
Course Name English:	**Game Design Fundamentals - From Concept to Creation**
Subject Code:	**GDC301**
Learning-Teaching Method:	Project based learning, In-class lecture, Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	45h (60 sessions) contact hours + 1h final exam + 104h self-study
Pre-Requisite:	PRO192
Description:	Embark on a journey into the world of game design with this comprehensive course. Before diving into coding, we'll explore the foundational elements that shape engaging video games. Discover the core concepts that drive player interaction, understand the mechanics and rules that define different game genres, and explore the art of storytelling in gaming. This course encourages you to think creatively and take risks as you develop your own unique game concepts. You'll learn to craft compelling narratives, design immersive environments, and create memorable characters. We'll also guide you through hands-on tutorials, enabling you to experiment with gameplay mechanics without the need for extensive coding. Key Learning Outcomes: - Understand and apply core game design frameworks (MDA, Core Loops, Design Pillars) to analyze existing games and design new mechanics - Design and document game systems including progression, economy, and balance using industry-standard GDD formats - Design player-centric levels that effectively guide, challenge, and teach players through intentional spatial layout and pacing - Design characters and combat systems with clear gameplay roles, readable animations, and balanced combat parameters - Craft compelling game narratives including premises, themes, and story structures that enhance player experience Integrate and direct game art assets to create cohesive visual experiences that support gameplay readability Implement game feel and physics-based interactions including responsive controls, animation states, and satisfying feedback - Execute complete game development cycle from concept through production to showcase using iterative design and playtesting methodologies By the end of this course, you'll have a solid foundation in game design, equipped to embark on your own game development journey.
StudentTasks:	- Students must attend more than 80% of contact slots in order to be accepted to the final examination. - Student is responsible to do all exercises, assignments and labs given by instructor in class or at home and submit on time - Use laptop in class only for learning purpose
Tools:	- Visual Studio 2019 or later (C# programming language) - Unity Hub and Unity 5 (version 2021.3 LTS or later) - ProBuilder (in Unity) - Git + GitHub (Student Pack) - Figma/Canva (web based) - Draw.io (web based) - Minimum hardware requirement: Windows 10, 64-bit versions, CPU: X64, RAM: 8GB - Students use Unity with their personal email for educational purposes only
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	1318/QĐ-ĐHFPT dated 11/27/2025
IsApproved:	**True**
Note:	
Is Scored:	**True**

MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	11/27/2025

4 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Level Up! The Guide to great video game design	Scott Rogers	Wiley	2024	3rd	978-1394298761	True	True	False	
2	Unity 6.2 User Manual	Unity	Unity				False	False	True	https://docs.unity3d.com/Manual/index.html
3	Unity Learn Platform	Unity	Unity				False	False	True	https://learn.unity.com
4	Game Design Workshop: A Playcentric Approach to Creating Innovative Games	Tracy Fullerton	A K Peters/CRC Press	2024	5th	978-1032607016	False	True	False	

8 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Understand and apply core game design frameworks (MDA, Core Loops, Design Pillars) to analyze existing games and design new mechanics
2	CLO2	Design and document game systems including progression, economy, and balance using industry-standard GDD formats
3	CLO3	Design player-centric levels that effectively guide, challenge, and teach players through intentional spatial layout and pacing
4	CLO4	Design characters and combat systems with clear gameplay roles, readable animations, and balanced combat parameters
5	CLO5	Craft compelling game narratives including premises, themes, and story structures that enhance player experience
6	CLO6	Integrate and direct game art assets to create cohesive visual experiences that support gameplay readability
7	CLO7	Implement game feel and physics-based interactions including responsive controls, animation states, and satisfying feedback
8	CLO8	Execute complete game development cycle from concept through production to showcase using iterative design and playtesting methodologies

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Course Overview & Assessment Module 1: Game Design Fundamentals 1.1 Aesthetics-Dynamics-Mechanics Framework	Offline	CLO1	IT	Slides Textbook	GDC301	Skim required readingfollow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project.	
2	1.2 ADM Analysis Workshop	Offline	CLO1	T	Slides Textbook		Collaborate in teams; run peer reviews or playtests; document decisions and next steps in the team log. Produce a one-page MDA analysis of a reference game.	
3	1.3 Core Loops & Design Pillars	Offline	CLO1	T	Slides Textbook		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Draft a 5–7 node core loop and define 2 KPIs (e.g., time-to-fun, completion%).	
4	1.4 Project Scoping & Risk Analysis	Offline	CLO1, CLO8	T	Slides Textbook		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project.	
5	1.5 Ideation Techniques Workshop ASSIGN PROJECT & ASSIGNMENT 1 INTRODUCTION	Offline	CLO1	TU	Slides		Collaborate in teams; run peer reviews or playtests; document decisions and next steps in the team log.	
6	Module 2: Prototyping & Early Documentation 2.1 Paper Prototyping Fundamentals	Offline	CLO1, CLO8	T	Slides		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Build a paper prototype; run 2 cross-tests; log time-to-fun and issues.	
7	2.2 Paper Prototyping Lab	Offline	CLO1, CLO8	TU	Slides Textbook		Follow the lecture (and/or demo); complete lab checkpoints; commit to Git with a short README and a GIF/video capture. Build a paper prototype; run 2 cross-tests; log time-to-fun and issues.	
8	2.3 Unity Graybox with ProBuilder	Offline	CLO1, CLO8	TU	Slides Textbook Paper prototyping quick guide		Follow the lecture (and/or demo); complete lab checkpoints; commit to Git with a short README and a GIF/video capture. Create the first room with spawn/camera and traversal; push to repo. Implement HUD elements and basic audio mix; quick usability check.	
9	2.4 GDD Structure Introduction	Offline	CLO2	T	Slides Unity Manual		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Fill One-sheet/Ten-pager sections; align with the pitch deck.	
10	2.5 Risk-First Prototyping	Offline	CLO1, CLO8	T	Slides Textbook		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project.	
11	2.6 GDD Draft Workshop	Offline	CLO1, CLO2	TU	Slides		Collaborate in teams; run peer reviews or playtests; document decisions and next steps in the team log. Fill One-sheet/Ten-pager sections; align with the pitch deck.	

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12	Module 3: Systems Foundation 3.1 Game Systems Overview ASSIGNMENT 2 INTRODUCTION SUBMIT ASSIGNMENT 1	Offline	CLO2	T	Slides Textbook		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Sketch the progression/economy loop and list incentives/sinks.	
13	3.2 Systems Design Workshop	Offline	CLO2	TU	Slides Textbook Sample spreadsheets (DPS/TTK)		Collaborate in teams; run peer reviews or playtests; document decisions and next steps in the team log. Sketch the progression/economy loop and list incentives/sinks.	
14	Module 4: Concept Pitch Preparation 4.1 Pitch Structure & Storytelling	Offline	CLO1, CLO2	T	Slides Textbook Sample spreadsheets (DPS/TTK)		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Fill One-sheet/Ten-pager sections; align with the pitch deck.	
15	4.2 Pitch Development Workshop 4.3 Demo Rehearsal & Feedback	Offline	CLO1, CLO8	TU	Slides Textbook		Collaborate in teams; run peer reviews or playtests; document decisions and next steps in the team log. Fill One-sheet/Ten-pager sections; align with the pitch deck.	
16	CONCEPT PITCH PRESENTATIONS	Offline	CLO1, CLO2, CLO8	U	Slides		Present within time limit; answer Q&A; submit deck/build and an iteration plan based on feedback. Fill One-sheet/Ten-pager sections; align with the pitch deck.	
17	Module 5: GDD Complete & Narrative 5.1 Premise, Theme & Player Experience 5.2 Story Beat Sheet & Pacing	Offline	CLO2, CLO5	T	Slides Textbook		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project.	
18	5.3 World Building Lab	Offline	CLO2, CLO5	TU	Slides		Follow the lecture (and/or demo); complete lab checkpoints; commit to Git with a short README and a GIF/video capture. Implement HUD elements and basic audio mix; quick usability check.	
19	5.4 GDD Vision, Pillars & Core Loop	Offline	CLO2	T	Slides Unity Manual		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Draft a 5–7 node core loop and define 2 KPIs (e.g., time-to-fun, completion%). Fill One-sheet/Ten-pager sections; align with the pitch deck.	
20	5.5 GDD Narrative & Technical Sections	Offline	CLO2, CLO5	TU	Slides		Follow the lecture (and/or demo); complete lab checkpoints; commit to Git with a short README and a GIF/video capture. Fill One-sheet/Ten-pager sections; align with the pitch deck.	
21	5.6 GDD v1 Peer Review	Offline	CLO2, CLO5	U	Slides Textbook		Collaborate in teams; run peer reviews or playtests; document decisions and next steps in the team log. Fill One-sheet/Ten-pager sections; align with the pitch deck. Prepare protocol/forms; run or participate in tests; analyze data; write postmortem.	

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22	Module 6: Level Design 6.1 Player Guidance Principles	Offline	CLO3	T	Slides Textbook Unity Analytics		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Implement HUD elements and basic audio mix; quick usability check.	
23	6.2 Gating & Challenge Pacing	Offline	CLO3	T	Slides Textbook Unity Manual		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Apply Three Cs; place landmarks and gates; record a 30–60s walkthrough.	
24	6.3 Level Blockout Theory	Offline	CLO3	T	Slides Textbook		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Apply Three Cs; place landmarks and gates; record a 30–60s walkthrough.	
25	6.4 Level Blockout Lab - Part 1 6.5 Level Blockout Lab - Part 2	Offline	CLO3	TU	Slides Unity Manual		Follow the lecture (and/or demo); complete lab checkpoints; commit to Git with a short README and a GIF/video capture. Apply Three Cs; place landmarks and gates; record a 30–60s walkthrough.	
26	6.6 Goals, Checkpoints & Camera	Offline	CLO3	TU	Slides Unity Manual		Follow the lecture (and/or demo); complete lab checkpoints; commit to Git with a short README and a GIF/video capture. Apply Three Cs; place landmarks and gates; record a 30–60s walkthrough.	
27	6.7 Onboarding & Playtest Prep	Offline	CLO3	TU	Slides Textbook		Collaborate in teams; run peer reviews or playtests; document decisions and next steps in the team log. Apply Three Cs; place landmarks and gates; record a 30–60s walkthrough. Prepare protocol/forms; run or participate in tests; analyze data; write postmortem (for S60).	
28	Progress Test 1	Offline	CLO1, CLO2, CLO3, CLO5, CLO8	U	Progress Test 1		Progress Test 1	
29	Module 7: System Balancing 7.1 Balancing: Difficulty Curves SUBMIT ASSIGNMENT 2 ASSIGNMENT 3 INTRODUCTION	Offline	CLO2, CLO8	T	Slides Textbook Unity Analytics		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Create/adjust a balancing spreadsheet; plan one A/B variable.	
30	7.2 Spreadsheet Balancing	Offline	CLO2, CLO8	T	Slides Textbook		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Create/adjust a balancing spreadsheet; plan one A/B variable.	
31	7.3 Parameterization & Tuning	Offline	CLO2, CLO8	T	Slides Textbook Sample spreadsheets (DPS/TTK)		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Create/adjust a balancing spreadsheet; plan one A/B variable.	

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32	7.4 Balance Workshop	Offline	CLO2, CLO8	U	Slides Textbook Sample spreadsheets (DPS/TTK)		Follow the lecture (and/or demo); complete lab checkpoints; commit to Git with a short README and a GIF/video capture. Create/adjust a balancing spreadsheet; plan one A/B variable.	
33	7.5 Progression & Economy Design	Offline	CLO2, CLO8	T	Slides Textbook Sample spreadsheets (DPS/TTK)		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Sketch the progression/economy loop and list incentives/sinks.	
34	Module 8: Vertical Slice Planning 8.1 Vertical Slice Scope & Criteria	Offline	CLO2, CLO8	T	Slides Textbook Sample spreadsheets (DPS/TTK)		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Define acceptance criteria and a cut-list aligned to the sprint.	
35	8.2 Vertical Slice Integration Sprint	Offline	CLO8	TU	Slides		Collaborate in teams; run peer reviews or playtests; document decisions and next steps in the team log. Define acceptance criteria and a cut-list aligned to the sprint.	
36	Module 9: Character & Combat Design 9.1 Hero Design: Silhouette & Role	Offline	CLO4	T	Slides		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Define roles and telegraphs; capture GIFs of readable attacks/counters.	
37	9.2 Enemy Archetypes & Telegraphing 9.3 Combat Feel: Anticipation-Action-Recovery	Offline	CLO4, CLO7	T	Slides Textbook		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Define roles and telegraphs; capture GIFs of readable attacks/counters.	
38	9.4 Animator: Locomotion States	Offline	CLO4, CLO7	TU	Slides Textbook		Follow the lecture (and/or demo); complete lab checkpoints; commit to Git with a short README and a GIF/video capture. Build state machines; set transitions; verify exit times and blends.	
39	9.5 Animator: Combat States	Offline	CLO4, CLO7	TU	Slides Unity Manual		Follow the lecture (and/or demo); complete lab checkpoints; commit to Git with a short README and a GIF/video capture. Define roles and telegraphs; capture GIFs of readable attacks/counters. Build state machines; set transitions; verify exit times and blends.	
40	9.6 Hitboxes, Feedback & Playtest Submit assignment 3	Offline	CLO4, CLO7	TU	Slides Textbook Unity Manual		Complete lab checkpoints, then run a short peer playtest; capture GIFs/logs and record issues. Implement hit/hurtboxes and i-frames; run a 5-player playtest and collect TTK metrics. Prepare protocol/forms; run or participate in tests; analyze data; write postmortem (for S60).	
41	Module 10: Art Integration 10.1 Art Direction for Designers	Offline	CLO6	T	Slides Textbook Unity Analytics		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Curate coherent assets; import & set materials/LODs; maintain readability.	

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42	10.2 Unity Import Pipeline	Offline	CLO6	TU	Slides Unity Manual		Follow the lecture (and/or demo); complete lab checkpoints; commit to Git with a short README and a GIF/video capture. Curate coherent assets; import & set materials/LODs; maintain readability.	
43	10.3 Art Integration Lab	Offline	CLO6	U	Slides Unity Manual		Follow the lecture (and/or demo); complete lab checkpoints; commit to Git with a short README and a GIF/video capture. Curate coherent assets; import & set materials/LODs; maintain readability.	
44	MIDTERM DEMO: Vertical Slice Review	Offline	All	U	Slides Textbook Unity Manual		Show the current build; receive targeted feedback; update the issue tracker with prioritized fixes. Define acceptance criteria and a cut-list aligned to the sprint. Prepare protocol/forms; run or participate in tests; analyze data; write postmortem (for S60).	
45	Module 11: Physics, Feel & UI 11.1 Movement Controllers & Physics	Offline	CLO7	T	Slides Textbook Unity Analytics		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Implement a controller; test collisions and slopes; document trade-offs.	
46	11.2 Physics & Custom Gravity Lab	Offline	CLO7	TU	Slides Unity Manual		Follow the lecture (and/or demo); complete lab checkpoints; commit to Git with a short README and a GIF/video capture. Implement a controller; test collisions and slopes; document trade-offs.	
47	11.3 Input Buffering & Juice	Offline	CLO7	TU	Slides Unity Manual		Collaborate in teams; run peer reviews or playtests; document decisions and next steps in the team log. Add buffering (50–150ms), screen shake, particles, and SFX timing; record before/after clips. Implement HUD elements and basic audio mix; quick usability check.	
48	11.4 HUD & UI Design Lab	Offline	CLO3, CLO6	TU	Slides Unity Manual		Follow the lecture (and/or demo); complete lab checkpoints; commit to Git with a short README and a GIF/video capture. Implement HUD elements and basic audio mix; quick usability check.	
49	11.5 Audio Integration Lab	Offline	CLO3	TU	Slides Unity Manual		Follow the lecture (and/or demo); complete lab checkpoints; commit to Git with a short README and a GIF/video capture. Implement HUD elements and basic audio mix; quick usability check.	
50	11.6 First Playable Build Workshop	Offline	CLO8	TU	Slides Unity Manual		Collaborate in teams; run peer reviews or playtests; document decisions and next steps in the team log. Implement HUD elements and basic audio mix; quick usability check.	
51	Module 12: Playtesting Cycle 12.1 Onboarding & Accessibility	Offline	CLO3, CLO8	T	Slides Unity Manual		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Apply Three Cs; place landmarks and gates; record a 30–60s walkthrough. Prepare protocol/forms; run or participate in tests; analyze data; write postmortem (for S60).	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
52	12.2 Playtest Planning Workshop	Offline	CLO3, CLO8	TU	Slides Textbook Unity Analytics		Collaborate in teams; run peer reviews or playtests; document decisions and next steps in the team log. Prepare protocol/forms; run or participate in tests; analyze data; write postmortem (for S60).	
53	12.3 Cross-Team Playtest Session	Offline	CLO3, CLO8	TU	Slides Textbook Unity Analytics		Collaborate in teams; run peer reviews or playtests; document decisions and next steps in the team log. Prepare protocol/forms; run or participate in tests; analyze data; write postmortem (for S60).	
54	12.4 Data Analysis & Beta Readiness	Offline	CLO3, CLO8	TU	Slides Textbook Unity Analytics		Skim required reading; follow the lecture; complete a 10–15 min micro-exercise mapping the concept to your project. Prepare protocol/forms; run or participate in tests; analyze data; write postmortem (for S60).	
55	Progress test 2	Offline	CLO2, CLO3, CLO4, CLO6, CLO7, CLO8	U	Progress test 2		Progress test 2	
56	Final Polish & Review	Offline	CLO8	TU	Slides Textbook Unity Analytics		Show the current build; receive targeted feedback; update the issue tracker with prioritized fixes. Prepare protocol/forms; run or participate in tests; analyze data; write postmortem (for S60).	
57	FINAL SHOWCASE & POSTMORTEM	Offline	All	U	Slides Textbook Unity Analytics		Present within time limit; answer Q&A; submit deck/build and an iteration plan based on feedback. Prepare protocol/forms; run or participate in tests; analyze data; write postmortem (for S60).	
58	FINAL SHOWCASE & POSTMORTEM	Offline	All	U	Slides Textbook Unity Analytics		Present within time limit; answer Q&A; submit deck/build and an iteration plan based on feedback. Prepare protocol/forms; run or participate in tests; analyze data; write postmortem (for S60).	
59	FINAL SHOWCASE & POSTMORTEM	Offline	All	U	Slides Textbook Unity Analytics		Present within time limit; answer Q&A; submit deck/build and an iteration plan based on feedback. Prepare protocol/forms; run or participate in tests; analyze data; write postmortem (for S60).	
60	FINAL SHOWCASE & POSTMORTEM	Offline	All	U	Slides Textbook Unity Analytics		Present within time limit; answer Q&A; submit deck/build and an iteration plan based on feedback. Prepare protocol/forms; run or participate in tests; analyze data; write postmortem (for S60).	

0 Constructive question(s)

7 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment 1 (Milestone 1 of Project)	on-going	1	10.0%	>0	1-3 weeks		Project-based process products		Cover content 1,2	- Guided by instructor - Application - Work in group	Instructor has responsibility to review for students after graded.
2	Assignment 2 (Milestone 2 of Project)	on-going	1	10.0%	>0	1-3 weeks		Project-based process products		Cover content 3, 4, 5	- Guided by instructor - Application - Work in group	Instructor has responsibility to review for students after graded.
3	Assignment 3 (Milestone 3 of Project)	on-going	1	10.0%	>0	1-3 weeks		Project-based process products		Cover content 6, 7, 8, 9	- Guided by instructor - Application - Work in group	Instructor has responsibility to review for students after graded.
4	Progress test 1	on-going	1	5.0%	>0	30'		Multiple choices Marked by Computer or a suitable format	25	Cover content 1,2,3,4, 5, 6	in class, by instructor	Progress test must be taken right after the last lectures of required material. Instructor has responsibility to review the test for students after graded.
5	Progress test 2	on-going	1	5.0%	>0	30'		Multiple choices Marked by Computer or a suitable format	25	Cover content 7, 8, 9, 10, 11, 12	in class, by instructor	Progress test must be taken right after the last lectures of required material. Instructor has responsibility to review the test for students after graded.
6	Project	on-going	1	30.0%	>0	8-10 weeks		Project		All modules, chapters	- Guided by instructor - Application - Work in group	
7	Final exam	Final exam	1	30.0%	4	60'		Multiple choices Marked by Computer	50	concepts, algorithms; all studied chapters	by exam board, using computer	The exam questions must be updated or different at least 70% to the previous ones.